An Ai-powered Web-based Personal Finance Tracker with a Financial Advisor for University Students

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Declaration

We declare that this project has not been submitted to any other University for the award of a Degree in Informatics and Computer Sciences.

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Abstract

University students worldwide face mounting financial pressures that require careful management of limited and irregular income against recurring expenses such as rent, food, transport, and academic costs. Although digital budgeting tools are widely available, their effective use depends on a level of financial literacy that many students have yet to develop. As a result, students often record transactions without understanding the patterns those transactions reveal, leaving the gap between financial information and financial guidance largely unaddressed. This shortfall contributes to weak saving habits, avoidable debt, and persistent financial stress among the student population.

The current landscape of personal finance applications offers either tracking features without meaningful interpretation or generic advice that is not tailored to the realities of student life. Few platforms combine transaction logging, multi-wallet management, and goal tracking with personalised financial guidance grounded in the user's actual spending behaviour. This disconnect between data capture and actionable insight motivates the development of a more integrated solution.

This project proposes a web-based Personal Finance Tracker integrated with an AI-powered Financial Advisor, designed using Object-Oriented Analysis and Design and developed through the Agile methodology. The platform will allow users to manage multiple wallets, log income and expenses, perform wallet-to-wallet transfers, and set savings goals, while computing spending statistics and category breakdowns from transaction history. A Large Language Model accessed through an Application Programming Interface will interpret these insights alongside user-stated goals to deliver personalised, conversational financial guidance. The system will be built using modern web technologies to ensure accessibility, scalability, and a responsive user experience for students across diverse devices and contexts.

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Abbreviations

AI – Artificial Intelligence

API – Application Programming Interface

BICS – Bachelor of Informatics and Computer Science

CSS – Cascading Style Sheets

ERD – Entity Relationship Diagram

ETF – Exchange-Traded Fund

HTML – HyperText Markup Language

LLM – Large Language Model

OECD – Organisation for Economic Co-operation and Development

OOAD – Object-Oriented Analysis and Design

SCES – School of Computing and Engineering Sciences

SU – Strathmore University

UI – User Interface

UML – Unified Modeling Language

UX – User Experience

YNAB – You Need A Budget

Chapter 1: Introduction

1.1 Background

The management of personal finances has emerged as a critical life skill for university students, who must navigate a financial landscape characterised by limited and often irregular income from family allowances, scholarships, or part-time work, alongside recurring expenses such as accommodation, food, transport, and academic materials. The ability to make sound financial decisions during this formative period has lasting effects on financial wellbeing in adulthood, with stronger financial behaviour during early adulthood significantly predicting long-term financial wellbeing (Sabri et al., 2023).

Globally, levels of financial literacy among adults remain low. The OECD/INFE 2023 International Survey of Adult Financial Literacy, which covered 39 countries and economies, found that only 34 percent of adults across all participating countries reached the minimum target score for financial literacy, with even lower scores in many emerging economies (OECD, 2023). For students, the consequences of this gap are particularly acute because they must make consequential financial choices, including managing daily budgets, taking on debt, and beginning to save, often without the benefit of formal financial education. Empirical work in West Africa has reinforced these concerns: a study of 400 University of Ghana students found that respondents exhibited limited knowledge in core areas including interest compounding, the time value of money, inflation, and investment risk diversification (Blay et al., 2024).

In sub-Saharan Africa, the situation is more pronounced. Financial literacy in the region has been estimated at 32 percent, compared with 52 percent in high-income countries, with only 38 percent of Kenyans, 34 percent of Ugandans, and 40 percent of Tanzanians considered financially literate (Koomson et al., 2023). At the same time, East Africa has experienced rapid digital financial transformation, driven in large part by mobile money. M-PESA in Kenya has grown into a national financial backbone, with annual transaction values reaching 7.2 trillion Kenyan shillings by 2023, equivalent to roughly 55 percent of Kenya's gross domestic product (Fintech Association of Kenya, 2025). For students, this digital shift has produced a paradox: while financial services are more

accessible than ever, finances are now scattered across multiple channels, including bank accounts, mobile wallets, and cash, making consolidated tracking and informed decision-making more difficult.

In response to these challenges, a wide range of personal finance applications has emerged. Research on such tools has shown that they can produce measurable improvements in financial knowledge, attitudes, and behaviours, including better tracking of income and expenditure and greater resilience to financial shocks (French et al., 2021). However, these tools largely focus on transaction capture and category visualisation; they generate data but offer little interpretation, leaving users to translate spending records into actionable decisions on their own. Recent advances in artificial intelligence offer a promising path forward. Studies evaluating leading Large Language Models (LLMs) on standardised personal finance assessments report average accuracy rates of approximately 70 percent, with the most advanced models exceeding this threshold across multiple topics (Hean et al., 2025). Furthermore, LLMs offer a conversational capability that has the potential to substantially improve the quality of digital financial guidance and reduce the human interaction historically required for personalised advice (Lo & Ross, 2024).

Despite these advances, few existing platforms combine comprehensive transaction tracking, multi-wallet management, and goal setting with an integrated AI advisor that can interpret a user's data and respond conversationally. This gap is particularly evident in the student context, where the user base has unique financial circumstances, such as irregular income and academic-cost-dominated expenditure, that are not adequately served by tools designed for working professionals. This study proposes a web-based Personal Finance Tracker integrated with an AI-powered Financial Advisor for university students, designed to bridge the gap between financial information and financial guidance by interpreting transaction histories and stated goals through a conversational, context-aware interface.

1.2 Problem Statement

University students struggle to track daily expenses, manage finances spread across multiple wallets, and plan for future financial needs. With income arriving irregularly from allowances, scholarships, or part-time work, and with expenditure scattered across cash, mobile money platforms such as M-PESA, and bank accounts, students rarely maintain a consolidated view of their finances (Fintech Association of Kenya, 2025). This fragmentation leads to poor financial

decisions, missed savings targets, and avoidable indebtedness, with weak financial behaviour during early adulthood having documented downstream effects on long-term financial wellbeing (Sabri et al., 2023).

Existing personal finance applications fail to address this problem effectively because they focus on data capture without meaningful interpretation. Students can record transactions but receive little guidance on what those records reveal or how to act on them, and such tools only improve financial behaviour for users who already possess the financial literacy needed to interpret the outputs (French et al., 2021). For many students, that literacy has not been established, with global surveys reporting that only 34 percent of adults reach the minimum target score for financial literacy and with lower rates documented across sub-Saharan Africa, including 38 percent in Kenya (OECD, 2023; Koomson et al., 2023). The result is that data is captured but rarely translated into action.

Where guidance is offered by existing tools, it is typically generic, drawn from rules designed for working adults rather than from the realities of student life, where income is irregular and expenses are dominated by non-discretionary academic and living costs. This leaves students dependent on intuition or informal advice from family and peers, neither of which accounts for the user's specific circumstances. The combined effect is that students continue to experience financial stress, accumulate avoidable debt, and develop weak saving habits during a formative period whose financial behaviours shape adult financial wellbeing (Sabri et al., 2023; Blay et al., 2024).

1.3 Objectives

This subsection presents the general objective and the specific objectives that guide this study, followed by the corresponding research questions.

1.3.1 General Objective

To develop a web-based Personal Finance Tracker integrated with an AI-powered Financial Advisor for university students.

1.3.2 Specific Objectives

The specific objectives of this study are:

i. To investigate the current methods and tools used by university students to manage their personal finances.

ii. To identify the challenges students face in tracking expenses, managing multiple wallets, and setting savings goals using existing approaches.

iii. To design and develop a web-based Personal Finance Tracker integrated with an AI-powered Financial Advisor for university students.

iv. To test the developed system to evaluate its functionality, usability, and the quality of its AI-generated financial guidance.

1.3.3 Research Questions

The research questions arising from the specific objectives are:

i. What are the current methods and tools used by university students to manage their personal finances?

ii. What challenges do students face in tracking expenses, managing multiple wallets, and setting savings goals using existing approaches?

iii. How will a web-based Personal Finance Tracker integrated with an AI-powered Financial Advisor be designed and developed for university students?

iv. How will the developed system be tested to evaluate its functionality, usability, and the quality of its AI-generated financial guidance?

1.4 Justification

The proposed system is justified on academic, social, and technological grounds. Academically, financial literacy has been consistently identified as a critical determinant of student wellbeing and long-term financial outcomes. Empirical research has shown that financial behaviour during early adulthood meaningfully shapes financial wellbeing later in life (Sabri et al., 2023), yet university students continue to demonstrate limited knowledge in core financial concepts (Blay et al., 2024). By coupling transaction tracking with personalised conversational guidance, the proposed solution addresses both the informational and the interpretive dimensions of financial literacy in a single platform.

Socially, university students represent a population that is uniquely positioned to benefit from accessible, personalised financial guidance. Students operate on tight budgets, face unpredictable income streams, and form the financial habits that will shape their adult financial lives. In sub-

Saharan Africa, where financial literacy lags behind high-income regions and where mobile money has reshaped how money is held and moved (Koomson et al., 2023; Fintech Association of Kenya, 2025), a tool tailored to the student context has clear potential to reduce financial stress and encourage saving behaviour during a formative period. The web-based delivery model further ensures broad accessibility, requiring only an internet-connected device rather than a specific operating system or the latest hardware.

Technologically, recent advances in Large Language Models have made it feasible to deliver high-quality, conversational financial guidance at scale and at low cost (Hean et al., 2025). Lo and Ross (2024) argue that the conversational capability of LLMs may substantially improve the quality of digital financial advice and reduce the human interaction traditionally required for personalised guidance, while also identifying open challenges around domain expertise, trustworthiness, and regulatory conformance. By accessing a leading LLM through an Application Programming Interface and constraining the model to the user's transaction history and stated goals, the proposed system can offer guidance comparable to that of a generic human advisor without the prohibitive cost that has historically excluded students from such services. Research on existing personal finance applications has further shown that such tools, when actively used, produce measurable improvements in financial knowledge, attitudes, and behaviours (French et al., 2021), providing empirical grounding for the proposed approach.

1.5 Scope

This study focuses on the design, development, and testing of a web-based personal finance tracking platform with an integrated AI-powered financial advisor for university students. The platform will support user account creation and authentication, the management of multiple wallets representing different sources of funds, the logging of income and expense transactions, wallet-to-wallet transfers, and the setting and tracking of savings goals. The system will compute spending statistics and category breakdowns from the user's transaction history and use these insights to inform the AI advisor's responses. The advisor will be implemented by integrating a third-party Large Language Model through an Application Programming Interface, with prompts engineered to constrain responses to the user's financial context and stated goals.

The project will be delivered as a web application accessible via standard modern browsers on desktop and mobile devices. The target user base is university students, although the design will

not be restricted to any specific institution. Features outside the scope of this project include direct integration with bank or mobile money APIs, automated transaction import, payment processing, investment portfolio management, and tax filing. These represent natural extensions for future work but are deliberately excluded from this initial phase to ensure the core feature set is delivered to a high standard within the available time.

1.6 Limitations

Several limitations are anticipated. First, the system will depend on users manually logging their transactions, which introduces the possibility of incomplete or inaccurate records that may degrade the quality of the AI advisor's recommendations. Second, access to the system requires an internet-connected device and a stable connection, which may not be uniformly available to all students. Third, the use of a third-party Large Language Model introduces a dependency on the availability, pricing, and policies of the chosen API provider, as well as the inherent variability of model outputs noted in recent evaluations of LLM-based financial guidance (Hean et al., 2025). Fourth, the AI advisor's guidance, while personalised, will not constitute regulated financial advice and may occasionally produce responses that are factually incorrect or contextually inappropriate, a known limitation of current generative artificial intelligence (Lo & Ross, 2024); mitigations such as constrained prompting and clear in-app disclaimers will be applied, but residual risk cannot be eliminated. Finally, user adoption may be hindered by privacy concerns associated with sharing financial data, particularly among students less familiar with cloud-based tools.

1.7 Delimitations

To keep the project tractable within the available time and resources, several deliberate boundaries have been set. The system will be developed exclusively as a web application, with native mobile applications excluded from the current scope. Transaction entry will be manual, with bank and mobile money integrations excluded. The AI advisor will be limited to non-investment, non-tax financial guidance and will rely on a single chosen LLM provider rather than a multi-model architecture. Testing will be conducted with a sample of university students recruited through convenience sampling, rather than a large-scale representative study. These delimitations ensure that the project scope remains focused on the core problem of bridging the gap between financial information and financial guidance for the student population.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews the literature relevant to personal finance management among university students and the digital tools that have emerged to support it. The chapter begins by examining personal finance management as it is experienced by the student population, before turning to the current approaches and tools that students use to manage their money. It then identifies the persistent challenges associated with these approaches and evaluates three leading AI-powered personal finance applications, namely Magnifi, You Need A Budget, and Cleo, in order to highlight the gaps that the proposed solution is designed to address. The chapter further reviews the technologies that the study draws upon, including Large Language Models, web application frameworks, and database systems, and concludes with the conceptual framework that guides the design of the proposed system.

2.2 Personal Finance Management Among University Students

Personal finance management refers to the process of planning, tracking, and controlling income and expenditure in order to meet short-term needs and longer-term goals. For university students, this process is shaped by a distinctive set of conditions: income that is typically irregular and modest, drawn from family allowances, scholarships, or part-time work, and expenditure that is dominated by non-discretionary academic and living costs. Empirical evidence shows that financial behaviour during this formative period has lasting consequences, with stronger financial behaviour during early adulthood significantly predicting financial wellbeing later in life (Sabri et al., 2023). The decisions students make about budgeting, spending, and saving therefore extend in importance beyond their immediate impact on day-to-day life.

The capacity of students to make sound financial decisions, however, is constrained by limited financial literacy. The OECD/INFE 2023 International Survey of Adult Financial Literacy, covering 39 countries and economies, reported that only 34 percent of adults across all participating countries reached the minimum target score for financial literacy, with even lower rates documented in many emerging economies (OECD, 2023). Among student populations specifically, the picture is consistent. A study of 400 University of Ghana students found that respondents demonstrated limited knowledge of core financial concepts including interest compounding, the time value of money, inflation, and investment risk diversification (Blay et al.,

2024). In sub-Saharan Africa more broadly, financial literacy has been estimated at 32 percent compared with 52 percent in high-income countries, with country-level rates of 38 percent in Kenya, 34 percent in Uganda, and 40 percent in Tanzania (Koomson et al., 2023). These figures point to a population of students who face real financial pressures while lacking the knowledge base required to navigate them confidently.

The financial environment in which African students operate has also undergone substantial transformation through the rapid adoption of mobile money. M-PESA in Kenya has grown from a peer-to-peer money transfer service into a national financial backbone, with annual transaction values reaching 7.2 trillion Kenyan shillings by 2023, equivalent to roughly 55 percent of Kenya's gross domestic product (Fintech Association of Kenya, 2025). For students, this transformation has produced a paradox. While digital financial services are more accessible than ever, students now hold and move money across multiple channels including bank accounts, mobile wallets, and cash. This fragmentation has made consolidated tracking and informed financial decision-making more difficult than in earlier eras when most transactions flowed through a single account.

2.3 Current Approaches to Personal Finance Management

University students currently rely on a mix of informal and digital approaches to manage their personal finances. The most common informal approach is mental tracking, in which students carry a rough sense of their balances, recent expenses, and upcoming obligations without committing them to any external record. A second informal approach is the use of paper notebooks or spreadsheets to log income and expenses, which provides a written record but requires significant manual effort and offers little in the way of analysis or reminders. These informal approaches are well documented in the financial behaviour literature, where they are often associated with reactive rather than proactive money management (Sabri et al., 2023).

On the digital side, students increasingly use a combination of bank mobile applications, mobile money applications such as M-PESA, and dedicated personal finance applications. Bank and mobile money applications provide transaction histories and balance information for individual accounts but do not aggregate across providers, meaning that a student with a bank account, an M-PESA wallet, and some cash must consult multiple sources to construct a complete view of their finances (Fintech Association of Kenya, 2025). Dedicated personal finance applications attempt to fill this gap by allowing users to log transactions, categorise them, and visualise spending

patterns, with research showing that such tools, when actively used, can produce measurable improvements in financial knowledge, attitudes, and behaviours including better tracking of income and expenditure and greater resilience to financial shocks (French et al., 2021). A new generation of these applications integrates artificial intelligence to deliver personalised guidance alongside tracking, with platforms such as Magnifi, You Need A Budget (YNAB), and Cleo representing distinct approaches that are reviewed in detail in section 2.5.

More recently, generative artificial intelligence has begun to feature in informal personal finance management. Students now ask Large Language Models such as ChatGPT, Claude, and Gemini for guidance on budgeting, saving, and financial planning (Mross, 2026). Evaluations of these models indicate that they typically offer acceptable answers for personal finance beginners, including college students, although their responses are often limited and rarely suggest seeking additional help from a professional (Mross, 2026). This emerging practice points to a clear demand for conversational financial guidance that existing personal finance applications have not yet met in an integrated way.

2.4 Challenges in Current Approaches

Despite the variety of approaches available, students continue to face several persistent challenges. The first is fragmentation. Because finances are spread across multiple channels, students must consult several applications and records to construct a complete picture of their position. This effort is repetitive and prone to omission, with the result that many students simply give up on consolidated tracking and fall back on intuition (Fintech Association of Kenya, 2025).

The second challenge is the gap between data and interpretation. Many existing personal finance applications focus on data capture and category visualisation, generating dashboards of where money has gone but offering limited guidance on what those records reveal or how to act on them. The benefits of these tools have been shown to depend on the user already possessing the financial literacy needed to interpret their outputs (French et al., 2021). For many students, that literacy has not been established (OECD, 2023; Blay et al., 2024; Koomson et al., 2023), with the result that data is captured but rarely translated into action.

The third challenge is the limited relevance of generic financial guidance. Where guidance is offered, it is typically drawn from rules designed for working adults with stable incomes, fixed

monthly bills, and long-term retirement horizons. Student life departs from these assumptions in important ways: income is irregular, expenses are dominated by non-discretionary academic and living costs, and savings horizons are short. Generic guidance therefore frequently misses the mark for the student user. Recent research on Large Language Models indicates that conversational artificial intelligence has reached a level of competence sufficient to deliver context-specific guidance (Hean et al., 2025; Lo & Ross, 2024), yet this capability has not been meaningfully integrated into existing student-focused tools.

The fourth challenge concerns geographic and pricing access. Several leading AI-powered personal finance applications are available only in selected high-income markets, such as the United States, United Kingdom, and Canada (Cleo AI, 2025; WallStreetZen, 2025), placing them out of reach for students elsewhere. Others, such as YNAB, are subscription-based and may be financially out of reach for many students despite educational discounts (CNBC Select, 2024). The combined effect is that students in markets such as Kenya have limited access to AI-powered personal finance tools tailored to their realities.

2.5 Related Applications

This section reviews three AI-powered personal finance applications that share elements of the proposed system's intended functionality. Magnifi represents the AI-powered investing assistant; You Need A Budget represents the methodology-driven, paid budgeting tool that has incorporated AI-assisted features; and Cleo represents the contemporary AI-powered conversational financial assistant. Together they cover the spectrum from investment-focused AI guidance through structured budgeting to free-form conversational coaching, and provide a useful reference against which to position the proposed solution.

2.5.1 Magnifi

Magnifi is an AI-powered investing platform developed by Magnifi LLC, a subsidiary of the financial technology firm TIFIN, founded in 2018 by Vinay Nair and Isabelle Freidheim and headquartered in the Greater Denver Area, United States (TheCollegeInvestor, 2025; WallStreetZen, 2025). The platform combines a conversational artificial intelligence assistant built on ChatGPT and proprietary Magnifi models with a marketplace of more than 15,000 stocks, exchange-traded funds, and mutual funds, allowing users to ask natural-language investment questions such as "companies investing in green energy" and receive ranked, data-backed results

(WallStreetZen, 2025). Users can also connect external brokerage accounts to obtain consolidated portfolio analytics, risk assessments, and personalised recommendations.

Magnifi offers a free tier that supports browsing, education, and commission-free trading, alongside a Magnifi Personal subscription priced at approximately 14 United States dollars per month or 132 United States dollars per year, which unlocks the AI investing assistant and advanced portfolio analytics (Benzinga, 2025; WallStreetZen, 2025). However, Magnifi is exclusively focused on investing rather than day-to-day budgeting, expense tracking, or multi-wallet management, and the platform is available only in the United States, United Kingdom, and Canada (WallStreetZen, 2025). Its AI guidance therefore addresses a different aspect of personal finance from the daily budgeting needs that dominate student life.

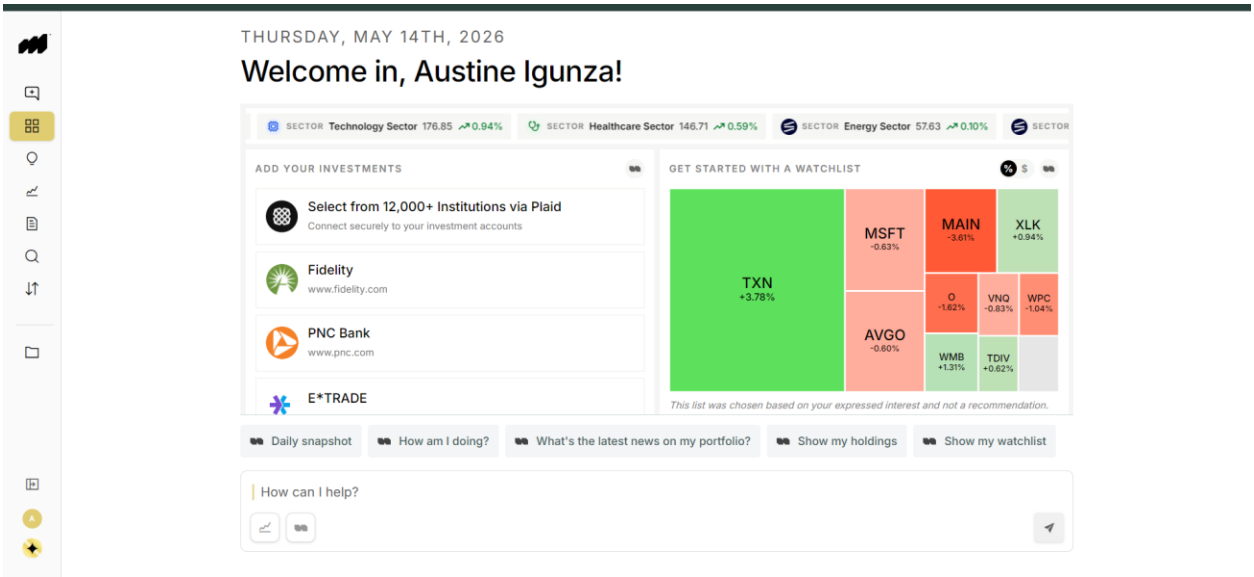


Figure 2.1: Magnifi AI-Powered Investing *Interface*

2.5.2 You Need A Budget (YNAB)

You Need A Budget, commonly abbreviated as YNAB, is a personal budgeting program originally developed in 2004 by Jesse Mecham, who built it as a spreadsheet to manage his own finances while pursuing a master's degree in accounting (YNAB, 2025). The platform transitioned to a software-as-a-service model in 2015 and is currently delivered through web and mobile applications. YNAB is built around a zero-based budgeting methodology summarised in four rules: give every dollar a job, embrace true expenses, roll with the punches, and age your money (YNAB,

2025). Users assign every unit of income to a specific category before spending it, with the aim of being intentional about every financial decision.

YNAB is a paid product priced at approximately 14.99 United States dollars per month or 109 United States dollars per year, with a 34-day free trial and a 12-month free subscription for students who provide proof of enrolment (CNBC Select, 2024). The platform is highly regarded for its educational orientation, offering structured tutorials and live workshops alongside its budgeting features, and is regularly named among the best budgeting applications by financial publications (CNBC Select, 2024). However, its strict zero-based methodology has a steep learning curve, and its subscription pricing places it out of reach for some students despite the educational discount.

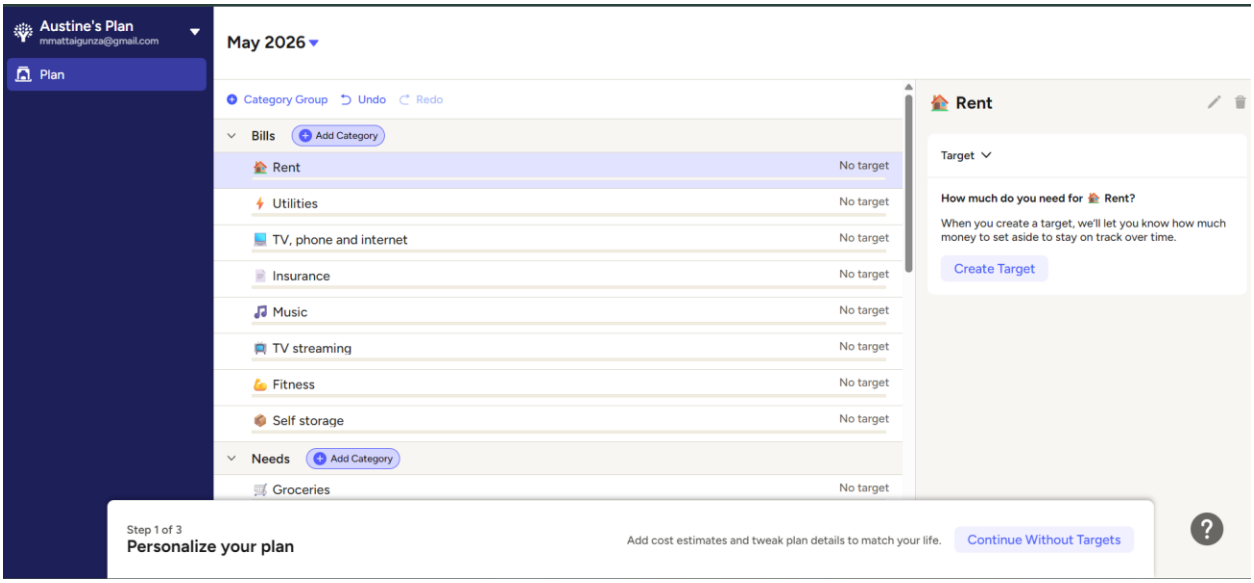


Figure 2.2: YNAB Zero-Based Budgeting Dashboard

2.5.3 Cleo

Cleo is an artificial-intelligence-powered financial assistant developed by Cleo AI, a London-based company founded in 2017 by Barney Hussey-Yeo (FinTech Magazine, 2024). The platform delivers personal finance management through a conversational chatbot interface that uses natural language processing to interpret users' spending patterns and respond with personalised advice in either a humorous or supportive tone (Cleo AI, 2025). Cleo connects to users' bank accounts to

provide real-time spending insights, set budgets, and run savings challenges, and the platform has been particularly successful with younger demographics, with the company reporting that users engage with Cleo approximately twenty times more than they do with typical banking applications (Cleo AI, 2025).

By 2025 Cleo reported more than one million paid subscribers and had analysed over 14 billion user transactions, generating an annual recurring revenue exceeding 250 million United States dollars (Cleo AI, 2025). However, Cleo has been available only in the United States since the company exited the United Kingdom market in 2020, and its conversational guidance is geared towards a general adult audience rather than the specific circumstances of university students. The platform also does not offer multi-wallet management at the granularity required by students juggling cash, mobile money, and bank accounts.

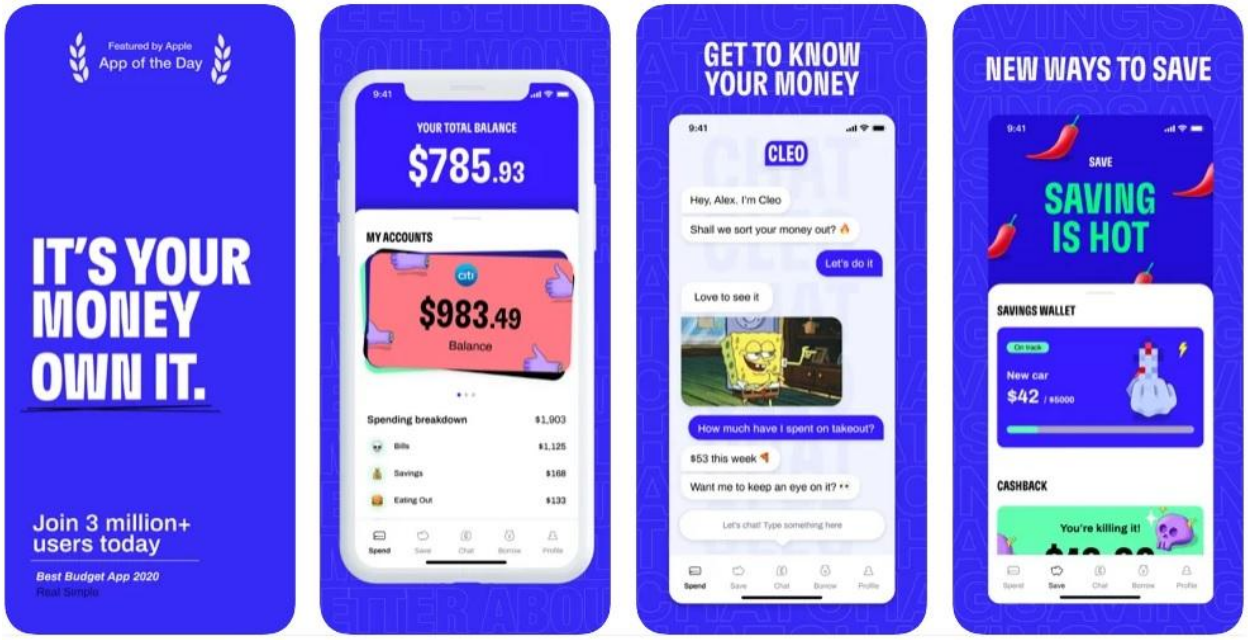


Figure 2.3: Cleo AI Conversational Financial Assistant Interface (Cleo AI, 2025)

2.6 Gaps in Current Approaches and Related Applications

Although Magnifi, YNAB, and Cleo each incorporate artificial intelligence to support personal finance management, none of them combines all of the elements that university students require. Magnifi offers a powerful conversational AI assistant but focuses exclusively on investing, leaving day-to-day budgeting, expense tracking, and multi-wallet management entirely outside its remit, and is restricted to users in the United States, United Kingdom, and Canada (WallStreetZen, 2025).

YNAB offers a rigorous, methodology-driven budgeting framework but assumes users have the time, motivation, and financial literacy to embrace zero-based budgeting, and its subscription pricing places a barrier in front of many students (CNBC Select, 2024). Cleo provides conversational AI guidance grounded in transaction data but is limited to the United States market, lacks support for the multi-wallet structure typical of student finances in markets such as Kenya, and is not specifically tuned to the constraints of student life (Cleo AI, 2025; FinTech Magazine, 2024).

Beyond the limitations of individual platforms, three structural gaps remain. First, no widely available platform combines unified transaction tracking, multi-wallet management, savings goal setting, and personalised conversational financial guidance in a single, integrated experience purpose-built for students. Second, existing AI-enabled tools are predominantly built for users in high-income markets and do not account for the multi-channel reality of mobile-money-driven economies in sub-Saharan Africa (Fintech Association of Kenya, 2025; Koomson et al., 2023). Third, the conversational guidance capability of contemporary Large Language Models, which has been shown to deliver context-specific personal finance advice with average accuracy rates of approximately 70 percent (Hean et al., 2025), has not been meaningfully integrated into student-focused personal finance applications that also handle the budgeting and tracking workflows specific to student life. The proposed system is positioned to address these gaps by combining comprehensive tracking, multi-wallet management, goal setting, and an integrated AI advisor that interprets the user's transaction history and stated goals through a conversational, context-aware interface.

2.7 Technologies Related to the Study

This section reviews the principal technologies that the study will draw upon. The selection focuses on four areas: Large Language Models, which power the AI advisor; web application frameworks, which deliver the user interface; server-side runtime environments, which host the application logic; and database systems, which store user data. Other technical decisions, such as deployment infrastructure and version control, are discussed in Chapter 3.

2.7.1 Large Language Models

A Large Language Model is a class of artificial intelligence model trained on extensive text datasets to generate and process natural language. Recent generations of LLMs, including

OpenAI's ChatGPT, Anthropic's Claude, Google's Gemini, and Meta's Llama, exhibit substantial capability in natural language understanding and content generation, and their outputs can be tailored to specific contexts through prompt engineering and the inclusion of structured input data (Hean et al., 2025). For personal finance applications, evaluations have reported average accuracy rates of approximately 70 percent across standardised personal finance assessments, with the most advanced models exceeding this threshold across multiple topics including credit cards, mortgages, taxes, and investments (Hean et al., 2025).

Lo and Ross (2024) argue that the conversational capability of LLMs has the potential to substantially improve the quality of digital financial guidance and reduce the human interaction historically required for personalised advice, while also identifying open challenges around domain expertise, trustworthiness, and regulatory conformance. Research aimed at college-aged users has further found that ChatGPT typically provides acceptable answers for personal finance beginners, although it rarely suggests seeking additional help from a professional (Mross, 2026). For the proposed system, an LLM will be accessed through an Application Programming Interface, with prompts engineered to constrain responses to the user's transaction history and stated goals, allowing the system to deliver guidance that is grounded in the user's actual financial behaviour.

2.7.2 React

React is a free and open-source JavaScript library for building user interfaces from reusable components, originally developed by Jordan Walke at Facebook and first deployed in production in 2011 before being open-sourced in 2013 (Meta Open Source, 2025). React is currently maintained by Meta and a community of contributors, and is one of the most widely used front-end technologies according to the Stack Overflow Developer Survey (Meta Open Source, 2025). The library uses a component-based architecture and a virtual Document Object Model that updates only the parts of the page that have changed, which contributes to improved rendering performance and developer productivity. React 19, released in December 2024, introduced server components and an improved compiler, further enhancing the library's suitability for modern web applications (Meta Open Source, 2025). React is appropriate for the proposed system because its component model supports the modular construction of dashboards, transaction lists, wallet views, and conversational interfaces.

2.7.3 Node.js

Node.js is an open-source, cross-platform JavaScript runtime built on Google's V8 engine that allows JavaScript code to be executed outside the browser, typically on the server (Node.js Foundation, 2025). Its non-blocking, event-driven architecture is well suited to handling many concurrent connections with low overhead, which makes it a common choice for web application back-ends, real-time services, and Application Programming Interface gateways. The use of JavaScript on both the client and the server simplifies development by allowing shared data structures, validation logic, and tooling across the stack. For the proposed system, Node.js will host the back-end services responsible for authentication, transaction storage, statistics computation, and orchestration of calls to the LLM provider's Application Programming Interface.

2.7.4 PostgreSQL

PostgreSQL is a free and open-source object-relational database management system with nearly four decades of active development (PostgreSQL Global Development Group, 2025). The system is fully compliant with the atomicity, consistency, isolation, and durability properties required for reliable transaction processing, and supports a rich set of data types including JSON, which combines the structural rigour of a relational database with the flexibility often associated with document stores. PostgreSQL is widely used in financial systems precisely because its transaction guarantees and point-in-time recovery capabilities support audit trails and regulatory compliance (PostgreSQL Global Development Group, 2025). For the proposed system, PostgreSQL will store users, wallets, transactions, savings goals, and conversation histories, providing a stable and queryable foundation upon which the spending statistics and category breakdowns that inform the AI advisor are computed.

2.8 Conceptual Framework

The conceptual framework of the proposed system describes how the principal actors and components interact. University students authenticate to the platform and use it to create wallets representing their cash, mobile money accounts, and bank accounts. Through the platform they log income and expense transactions, perform wallet-to-wallet transfers, and define savings goals. These data are persisted in a PostgreSQL database via a Node.js back-end. The back-end computes spending statistics, category breakdowns, and progress towards savings goals from the stored transactions, and exposes these as structured inputs to a prompt that is sent, alongside the student's

natural-language question, to a third-party Large Language Model through its Application Programming Interface. The model returns a personalised, conversational response that is rendered to the student through the React-based front-end. The conceptual framework is illustrated in Figure 2.4 below.

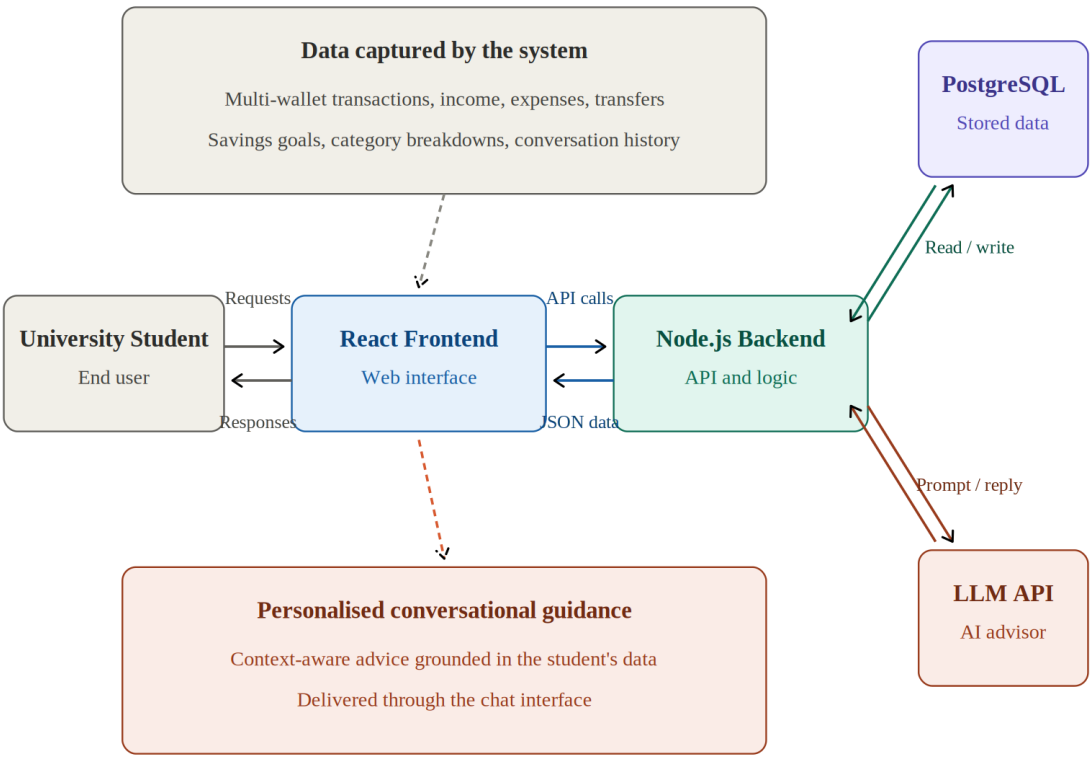


Figure 2.4: Conceptual Framework for the Proposed System

In this framework, the platform acts as the integration layer between the student's financial data and the AI advisor's interpretive capability. The wallet abstraction provides the multi-channel coverage that students require in mobile-money-driven economies, while the structured statistics passed to the LLM ensure that the advisor's responses are grounded in the user's actual financial behaviour rather than in generic rules. Together, these elements address the gaps identified in section 2.6 by combining unified tracking, multi-wallet management, goal setting, and personalised conversational guidance in a single integrated experience.

Chapter 3: Methodology

3.1 Introduction

This chapter presents the methodological framework that will guide the design, development, and evaluation of the proposed AI powered web-based Personal Finance Tracker with a Financial Advisor. The chapter begins by setting out the system development methodology selected for the project and the rationale for its choice. It then describes the analysis and design models, drawn from Object-Oriented Analysis and Design, that will be used to capture the structure and behaviour of the system. The chapter proceeds to outline the tools and technologies on which the system will be built, justifying each selection against the project objectives, and closes with the tangible deliverables that the project will produce.

3.2 Methodology

The project will adopt the Agile methodology, an iterative and incremental approach to software development that values flexibility, close collaboration with stakeholders, and the frequent delivery of working software (Beck et al., 2001; GeeksforGeeks, 2026). Rather than completing every requirement, design, and implementation activity for the entire system in one sequential pass, Agile organises the work into a series of short development cycles called iterations, with each iteration producing a working, demonstrable portion of the system that builds on what was delivered before (Hoda et al., 2021). Agile is grounded in the four values articulated in the Manifesto for Agile Software Development, which prioritise individuals and interactions, working software, customer collaboration, and responsiveness to change over rigid adherence to processes, documentation, and plans (Beck et al., 2001). The lifecycle followed by the project is illustrated in Figure 3.1 below.

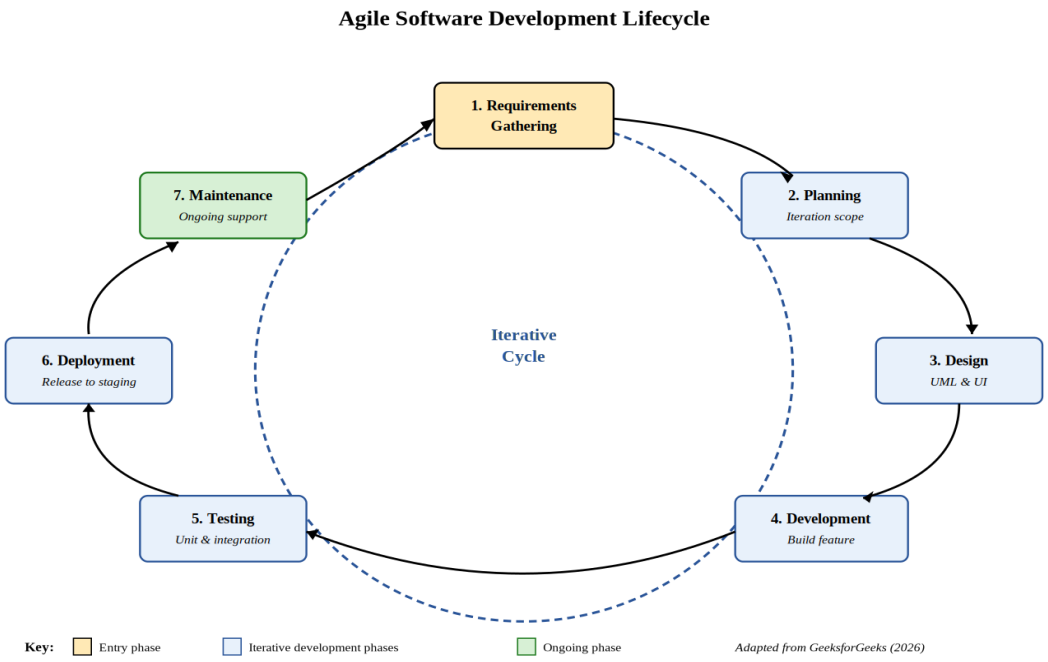


Figure 3.1: Agile Software Development Lifecycle

The subsections that follow describe the activities performed in each phase of the Agile lifecycle as applied to this project, after which a justification for selecting this methodology over alternative approaches such as pure Waterfall and Incremental Waterfall is provided.

3.2.1 Requirements Gathering

In this phase, the development team works to understand what the system must do for university students managing their personal finances. Requirements are elicited through reflection on the team's own needs as student users, review of related applications surveyed in Chapter 2, and engagement with the project supervisor, who guides the prioritisation of features. Once gathered, the requirements are sorted by priority so that the most important features, namely authentication, wallet management, and transaction logging, are developed before dependent features such as the AI advisor and analytics dashboard (GeeksforGeeks, 2026).

3.2.2 Planning

The planning phase translates the prioritised requirements into a concrete development plan. The team identifies the features to be delivered in the upcoming iteration, estimates the effort required,

and allocates time and resources accordingly (GeeksforGeeks, 2026). Planning is performed at the start of every iteration rather than once for the whole project, which allows priorities to be adjusted in response to what was learned in the previous iteration.

3.2.3 Design

The design phase translates the planned requirements into a concrete blueprint for implementation. Object-Oriented Analysis and Design techniques, described in section 3.3 and section 3.4, are used to model the static structure of the system through class and entity relationship diagrams and its dynamic behaviour through use case, sequence, system sequence, and activity diagrams. User interface wireframes and database schema updates are produced where the iteration introduces new screens or data entities. Design decisions are kept lightweight and refined as the team learns from each iteration, reflecting the Agile preference for evolving design over heavy upfront specification (Beck et al., 2001).

3.2.4 Development

During development, the design is realised in working source code. The team works in short, focused cycles, building small, usable pieces of the product, with each cycle building on the last so that quick feedback can be incorporated as the system grows (GeeksforGeeks, 2026). The front-end components are built in React, the back-end logic is developed in Node.js with the Express framework, and any database schema changes are applied to the PostgreSQL database. Code is committed regularly to a Git repository hosted on GitHub, with each commit accompanied by a clear message and reviewed against the iteration's goals before being merged.

3.2.5 Testing

As the software is built, it is tested to ensure it works properly and meets the requirements (. Unit tests are written for individual back-end functions and front-end components, integration tests verify the interactions between modules already delivered, and acceptance tests confirm that each feature satisfies its specified criteria. Testing happens continuously throughout each iteration so that defects are caught early, before they become bigger issues.

3.2.6 Deployment

Once an iteration's work has passed its testing gates, the working software is deployed so that it can be exercised end-to-end and reviewed against the requirements that drove the iteration . The project supervisor evaluates the deployed increment and provides direction on priorities for the

next iteration, while the team additionally trials the system on its own student finances to surface practical issues that pure inspection might miss.

3.2.7 Maintenance

Even after a feature is deployed, work on it does not end. The team continues to maintain delivered features as the system grows, ensuring that earlier work continues to function correctly as new features are integrated and that any changes in requirements or supervisor feedback are reflected in the evolving system . This is consistent with the Agile principle of welcoming changing requirements throughout the development lifecycle (Beck et al., 2001).

3.2.8 Justification of Methodology Choice

The Agile methodology was selected after careful consideration of the project context and the alternative methodologies available. Pure Waterfall was rejected because its strictly sequential phase structure postpones any working software until the end of the project, leaving no opportunity to surface design flaws, validate technology choices, or respond to supervisor feedback until very late in the lifecycle.

Agile, by contrast, is well suited to the project on two grounds. First, the proposed system decomposes naturally into user-facing capabilities, namely authentication, wallets, transactions, savings goals, the AI advisor, and the dashboard, that map cleanly onto iteration-sized deliveries. Second, the AI advisor component carries genuine technical uncertainty around prompt construction, model behaviour, and grounding in transaction data, and Agile's short feedback cycles allow these uncertainties to be probed empirically rather than committed to up front (Beck et al., 2001; Hoda et al., 2021).

3.3 Analysis

System analysis captures the requirements of the proposed system in models that describe what the system must do and how its data is structured. Following Object-Oriented Analysis and Design conventions, the analysis is performed through a set of standard Unified Modeling Language diagrams (Larman, 2004; Wazlawick, 2024). The subsections below describe the diagrams that will be produced; the diagrams themselves will be drawn and presented in Chapter 4.

3.3.1 Use Case Diagram

The use case diagram will identify the actors that interact with the system and the functional goals each actor pursues through it. The primary actor is the university student, who will create an account, manage wallets, log income and expense transactions, transfer funds between wallets, set savings goals, view spending statistics, and consult the AI advisor. A secondary actor is the AI advisor itself, treated as an external system accessed through the Anthropic Claude Application Programming Interface, which receives prompts containing the student's financial context and returns conversational responses. The use case diagram will support the elaboration of detailed use case descriptions used during the requirements phase of each increment.

3.3.2 Sequence Diagram

Sequence diagrams will model the time-ordered exchange of messages between the objects that collaborate to fulfil specific use cases. Diagrams will be produced for the central scenarios of the system, including logging a transaction, performing a wallet-to-wallet transfer, and requesting personalised guidance from the AI advisor. In the advisor scenario, the diagram will show how the front-end relays the student's question to the back-end, how the back-end constructs a context-aware prompt from the stored transaction history and savings goals, how that prompt is sent to the Claude Application Programming Interface, and how the response is returned and rendered to the student.

3.3.3 Sequence Diagram

System sequence diagrams will be produced as a complement to the use case descriptions. Where an ordinary sequence diagram models internal object interactions, a system sequence diagram treats the system as a single black box and shows the events that pass between the external actor and the system during a use case scenario (Larman, 2004). These diagrams will help establish the system interface and the data carried by each event before the internal object collaborations are designed.

3.3.4 Entity Relationship Diagram

The entity relationship diagram will model the persistent data of the system and the relationships between the principal entities, namely users, wallets, transactions, transaction categories, savings goals, and conversation messages. Cardinalities and key attributes will be represented, providing the basis from which the PostgreSQL database schema in section 3.4 will be derived.

3.4 Design

System design specifies how the system will satisfy the requirements captured during analysis. The design models described in this section express the static structure of the software, the dynamic flow of activity within key processes, and the visual and architectural framework in which these are realised. As with the analysis models, the diagrams themselves will be drawn and presented in Chapter 4.

3.4.1 Class Diagram

The class diagram will represent the principal classes of the system, their attributes and methods, and the structural relationships of association, aggregation, composition, and inheritance between them. Anticipated classes include User, Wallet, Transaction, Category, SavingsGoal, Conversation, and Message, together with controller classes that orchestrate transactional behaviour and a service class that mediates communication with the Claude Application Programming Interface. The class diagram will guide the implementation of the back-end domain model and the structure of the database schema (Larman, 2004; Wazlawick, 2024).

3.4.2 Activity Diagram

Activity diagrams will model the flow of control through the principal workflows of the system. Diagrams will be produced for the processes of logging a transaction, executing a wallet-to-wallet transfer, contributing to a savings goal, and conducting an advisory conversation with the AI advisor. Decision nodes will capture the validation rules enforced at each step, such as ensuring that a wallet contains sufficient funds before a transfer or that a transaction's amount and category are present and well formed.

3.4.3 Architecture

The system architecture will be expressed in a deployment-oriented diagram showing the principal tiers of the system and the technologies that occupy each tier. The front-end React application will run in the student's browser and will communicate over HTTPS with a Node.js back-end built on Express and deployed on Railway. The back-end will handle authentication, request authorisation, business logic, and orchestration of calls to the Anthropic Claude Application Programming Interface, and will persist user data in a PostgreSQL database also hosted on Railway. The front-end will be deployed on Vercel, which integrates with Git to provide automated builds and

previews. The architecture diagram will make the separation of concerns explicit and provide a reference for deployment activities in each increment.

3.4.4 Database Schema

The database schema will translate the entity relationship diagram into a concrete relational design suitable for PostgreSQL. Each entity will be represented as a table with appropriate primary and foreign keys, data types, integrity constraints, and indexes for fields that support frequent lookup, such as user identifiers on transaction tables. The schema will include audit fields such as creation and update timestamps and will be versioned through migration scripts to ensure that schema evolution between increments remains reproducible.

3.4.5 Wireframes

Wireframes will be produced for the principal screens of the application, including the registration and login screens, the main dashboard with spending summaries and goal progress, the wallets list and detail screens, the transaction entry form, the savings goals screen, and the AI advisor conversation interface. The wireframes will define layout, information hierarchy, and the placement of interactive elements, and will inform the implementation of the React components in each increment.

3.5 Development Tools and Technologies

This section presents the tools and technologies selected for the development of the proposed system. Each selection is described and justified against the requirements of the project, and a consolidated justification is presented in Table 3.1 at the end of the section.

3.5.1 Programming Language

JavaScript will serve as the primary programming language for both the front-end and the back-end of the system. JavaScript is one of the most widely used languages in modern web development and the use of a single language across both tiers of the application simplifies shared logic such as data validation and reduces the cognitive overhead of switching languages during development (Mozilla Developer Network, 2025).

3.5.2 Frontend Framework

The front-end will be built using React, a free and open-source JavaScript library for constructing user interfaces from reusable components (Meta Open Source, 2025). React's component-based

model is well suited to the modular structure of the proposed system, in which dashboards, wallet lists, transaction forms, and the AI advisor conversation can each be expressed as discrete components. React's virtual Document Object Model updates only the parts of a page that have changed, contributing to responsive interactions across desktop and mobile browsers.

3.5.3 Backend Runtime

Node.js will host the back-end services of the system. Node.js is a cross-platform JavaScript runtime built on the V8 engine, designed for non-blocking, event-driven server-side execution (Node.js Foundation, 2025). Its concurrency model handles many simultaneous requests efficiently, which is appropriate for the read-heavy access patterns of a finance application with many short interactions and occasional longer calls to the AI advisor.

3.5.4 Backend Framework

Express, a minimal and flexible web application framework for Node.js, will be used to build the back-end Application Programming Interface (Express, 2024). Express provides a lightweight routing and middleware system that supports the construction of clean, well-organised RESTful endpoints without imposing the overhead of a heavier framework. Express 5, released in October 2024, brings improved support for modern asynchronous patterns and serves as a stable base for the project (Express, 2024).

3.5.5 Database

PostgreSQL, a free and open-source object-relational database management system with nearly four decades of active development, will store the system's data (PostgreSQL Global Development Group, 2025). PostgreSQL is fully compliant with the atomicity, consistency, isolation, and durability properties required for reliable transaction processing, which is essential in a financial application where wallet-to-wallet transfers must be applied atomically to avoid inconsistent balances. Its support for JSON columns will also allow the system to persist conversation histories with the AI advisor in a flexible yet queryable form.

3.5.6 AI Service Provider

The AI-powered Financial Advisor component will be powered by Anthropic's Claude family of Large Language Models, accessed through the Claude Application Programming Interface (Anthropic, 2026). Claude was selected because the provider publishes detailed model documentation, capability cards, and safety evaluations that support a defensible engineering

choice, and because the Messages Application Programming Interface offers a straightforward conversational request-response pattern well suited to advisor interactions (Anthropic, 2026). The model accessed will be a current production Claude model; the back-end will construct prompts that include the student's transaction history, category breakdowns, and stated savings goals to ground the advisor's responses in the user's actual financial context (Lo & Ross, 2024).

3.5.7 Hosting and Deployment

The system will be deployed to two complementary cloud platforms. The React front-end will be hosted on Vercel, a platform purpose-built for modern front-end applications that provides automatic builds from Git, global content delivery, and preview deployments for every push (Vercel, 2026). The Node.js back-end and the PostgreSQL database will be hosted on Railway, an infrastructure platform that provides managed Node.js services and managed PostgreSQL within a single project, with environment-variable wiring between services (Railway, 2025). Both platforms offer free or low-cost tiers appropriate for a student project.

3.5.8 Styling and User Interface

Tailwind CSS, a utility-first cascading style sheet framework, will be used to style the front-end components (Tailwind Labs, 2025). Tailwind enables rapid, consistent styling directly within the React component markup, reducing the need for separate stylesheets and supporting a coherent visual language across the application.

3.5.9 Version Control and Collaboration

Git will be used for source code version control, with the project repository hosted on GitHub (GitHub, 2025). Git allows the development team to track every change, work in parallel through branches, and revert to earlier states when defects are introduced. GitHub provides a shared remote, pull-request based code review, issue tracking, and continuous integration capabilities that align well with the iterative delivery pattern of the chosen methodology.

3.5.10 Integrated Development Environment

Visual Studio Code will serve as the primary integrated development environment for the project (Microsoft, 2025). It offers strong native support for JavaScript, React, and Node.js, an extensive ecosystem of extensions for linting, formatting, and debugging, and integrated terminal and Git tooling that streamlines the development workflow.

Table 3.1: Summary of Development Tools and Technologies

Layer / Role	**Tool / Technology**	**Justification**
Programming language	JavaScript	Universally supported across front-end and back-end, removing the cognitive cost of cross-language development.
Frontend library	React	Component-based model aligns with the modular structure of the system; mature ecosystem and broad community support.
Backend runtime	Node.js	Event-driven, non-blocking I/O efficiently handles many concurrent requests typical of a finance application.
Backend framework	Express	Lightweight routing and middleware system suitable for building clean RESTful APIs without unnecessary complexity.
Database	PostgreSQL	ACID compliance is essential for reliable financial transactions; JSON support accommodates flexible advisor conversation data.
AI service	Anthropic Claude API	Strong conversational reasoning, transparent model documentation, and a straightforward Messages API for prompt-grounded guidance.
Frontend hosting	Vercel	Purpose-built for React deployments with Git integration, preview environments, and a free tier appropriate for student work.
Backend & DB hosting	Railway	Managed Node.js and managed PostgreSQL within a single project with built-in environment wiring; affordable for students.
Styling	Tailwind CSS	Utility-first styling that keeps presentation logic close to components and ensures visual consistency across the application.

Version control	Git and GitHub	Industry-standard version control with branch-based collaboration, code review, and integration with both Vercel and Railway.

3.6 Deliverables

The project will produce the following tangible deliverables that together constitute the complete output of the project scope. Each deliverable corresponds to one or more iterations of the Agile lifecycle described in section 3.2.

3.6.1 Project Proposal

The project proposal is a complete documentation of the rationale, requirements, design approach, methodology, technology selections, and expected deliverables of the project. It is the immediate deliverable of the proposal stage and the foundation against which all subsequent work will be assessed.

3.6.2 Authentication Module

The authentication module will provide secure user registration, login, password storage using industry-standard hashing, session management, and account recovery. It will be delivered in the first iteration because all other modules depend on the establishment of an authenticated user identity.

3.6.3 Wallet Management Module

The wallet management module will allow authenticated users to create, view, edit, and deactivate wallets representing distinct sources of funds, including cash, mobile money accounts, and bank accounts. Each wallet will display its current balance, with balances kept consistent with the transaction history stored in the system.

3.6.4 Transaction Management Module

The transaction management module will support the logging of income and expense transactions against the user's wallets, the categorisation of transactions, and the execution of wallet-to-wallet transfers as paired atomic operations. The module will also compute spending statistics and category breakdowns that feed the dashboard and the AI advisor.

3.6.5 Savings Goals Module

The savings goals module will allow users to define target amounts and deadlines for specific financial objectives, associate progress contributions with the goal, and track the proportion of the target achieved over time. Goal status will form one of the inputs passed to the AI advisor to inform its guidance.

3.6.6 AI Financial Advisor Module

The AI financial advisor module will provide a conversational interface through which users can ask questions about their finances and receive personalised responses generated by the Claude language model. The module will be responsible for constructing context-aware prompts that include the user's recent transaction history, category breakdowns, and active savings goals, for managing the conversational history, and for handling errors and rate-limiting from the upstream Application Programming Interface.

3.6.7 Dashboard and Analytics Module

The dashboard and analytics module will present aggregated views of the user's financial position, including total balances across wallets, spending by category over selectable time windows, income against expenditure trends, and progress towards active savings goals. It will act as the primary landing surface of the application after authentication.

3.6.8 Final Deployed System and Documentation

The final deliverable will be the complete integrated system, deployed and operational on Vercel and Railway, accompanied by the full project dissertation, user documentation, and the source code repository on GitHub. This deliverable represents the culmination of all preceding increments combined into a single coherent product.

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Appendix

Grantt Chart

